

Chapter 7

Introduction to Hyperfine Structure

The Fermi contact interaction for s states is covered in §6.1.1 of Foot [3].

7.1 Nuclear Spin

The atomic nucleus is like a little, linear, gas-phase molecule. The nucleons (*i.e.* protons and neutrons) have spin, but can also orbit around in the nucleus. They tend to arrange themselves into an axially-symmetric distribution that itself then rotates in the lab frame, and the resulting state can have nonvanishing expectation values for the even electric (and odd magnetic) multipole moments. The total angular momentum of the nucleus, which is a combination of spins and orbits, is referred to as “nuclear spin,” and the dimensionless quantum number I that quantifies the nuclear spin takes both integer and half-integer values.

Unlike the case for a fundamental particle, since the nucleus is a composite particle, each nucleus generally has a unique g factor that tells us the relationship between $\mathbf{I}$ and $\boldsymbol{\mu}$,

$$\boldsymbol{\mu}_I = \mathbf{I} g_I \mu_N \quad (7.1)$$

where the nuclear magneton is defined by

$$\mu_N \equiv \frac{e\hbar}{2m_p} \quad (7.2)$$

where $m_p \approx 1836 \times m_e$ is the proton rest mass. The magnitude of a typical nuclear magnetic moment, then, is expected to be about 2000 times smaller than a typical atomic magnetic moment.

The states of the nucleus in an atom could be expanded in a basis spanned by angular momentum eigenstates $\{|IM_I\rangle\}$. We will see, however, that the coupling between the electron angular momentum and the nuclear angular momentum will in most cases lead to energy eigenstates that are superpositions of states with different M_I and M_S (or, more generally, M_J) values, and the states are best described in terms of the total angular momentum $\mathbf{F}$.

7.2 Hyperfine interaction for hydrogenic electrons in s states

We will start with a derivation of the hyperfine interaction with s electrons (called the “Fermi contact interaction”) that considers the interaction between the *field* created by the electron distribution at the location of the nucleus with the *electromagnetic moments* of the nuclear charge distribution. We will then show that this same thing can be derived by instead considering the interaction between the *electromagnetic moments* of the electron distribution with the *field* created by the nucleus at the location of the electron.

7.2.1 The electron creates a magnetic field at the nucleus

For an electron in a state with $L = 0$ (which is to say an s state), the magnetic field at the nucleus can be attributed to the electron’s spin. In fact, we will argue that it is entirely the electron spin *evaluated at the position of the nucleus* that is responsible for this, as follows.

Classical electromagnetism can be used to show that the magnetic field at the center of a uniformly magnetized sphere of radius R is given by

$$\mathbf{B} = \frac{2\mu_0}{3}\mathbf{M} \quad (7.3)$$

where μ_0 is the permeability of free space.

The important thing to notice about this is that it doesn’t depend upon the radius R . So if we were to consider adding to one such sphere an outer,

hollow sphere to, say, create an effectively larger sphere with double the radius or whatever, the presence of this outer, hollow sphere of uniform magnetization would not change the field at the center¹. We can therefore conclude that a uniformly-magnetized spherical shell (even if its thickness is macroscopic) does not produce a magnetic field at its center.

From this argument, we see that the only part of the electron's spatial wavefunction that will contribute to the magnetic field at the nucleus is the value very near $r_e = 0$. (Note that $|\psi(r_e = 0)|^2$ is only nonzero for s states, so this interaction will be almost negligible for states with $L \neq 0$).

Equation (7.3) allows us to calculate the magnetic field at the origin if we know the magnetization at that location. If we were to write down the *operator* whose expectation value gives the magnetization at position $\mathbf{R}$ for an electron at position $\mathbf{r}_e$, it needs to be zero everywhere except at the position of the electron²,

$$\hat{\mathbf{M}}(\mathbf{R}, \mathbf{r}_e) = \boldsymbol{\mu}_S \delta(\mathbf{R} - \mathbf{r}_e). \quad (7.4)$$

The expectation value can be taken by integrating this over $d^3\mathbf{r}_e = d\phi_e d\theta_e dr_e \sin(\theta_e) r_e^2$ weighted by $|\psi_{n00}(\mathbf{r}_e)|^2$.

Since the spatial size of the nucleus is much smaller than the length scale over which the electron wavefunction varies (na_0), we can approximate the magnetization in the nucleus as taking a uniform value given by its expectation value at the origin,

$$\begin{aligned} \langle \hat{\mathbf{M}}(0, \mathbf{r}_e) \rangle &= \int d^3\mathbf{r}_e \psi_{n00}^*(\mathbf{r}_e) \langle \boldsymbol{\mu}_S \rangle \delta(0 - \mathbf{r}_e) \psi_{n00}(\mathbf{r}_e) \\ &= \langle \boldsymbol{\mu}_S \rangle \int_0^{2\pi} d\phi_e \int_0^\pi d\theta_e \sin(\theta_e) \int_0^\infty dr_e r_e^2 \frac{1}{4\pi r_e^2} \delta(r_e) |\psi_{n00}(r_e, \theta_e, \phi_e)|^2 \end{aligned} \quad (7.5)$$

$$= \frac{\langle \boldsymbol{\mu}_S \rangle}{4\pi} \int_0^{2\pi} d\phi_e \int_0^\pi d\theta_e \sin(\theta_e) |\psi_{n00}(0, \theta_e, \phi_e)|^2 \quad (7.6)$$

¹This is discussed in problem 2.5 in [1].

²The 3D Dirac delta function in spherical coordinates is related to 1D Dirac delta functions via

$$\delta(\mathbf{r} - \mathbf{r}_0) = \begin{cases} \frac{1}{r^2 \sin(\theta)} \delta(r - r_0) \delta(\theta - \theta_0) \delta(\phi - \phi_0) & \text{if } r_0 \neq 0 \text{ and } \theta_0 \text{ and } \phi_0 \text{ both defined} \\ \frac{1}{2\pi r^2 \sin(\theta)} \delta(r - r_0) \delta(\theta - \theta_0) & \text{if } r_0 \neq 0 \text{ and } \theta_0 = 0, \pi \text{ } (\phi_0 \text{ undefined}) \\ \frac{1}{4\pi r^2} \delta(r) & \text{if } r_0 = 0 \text{ (so } \theta_0 \text{ and } \phi_0 \text{ undefined)} \end{cases} .$$

Noting that if $r_e = 0$, any function of $\mathbf{r}_e$ cannot have any dependence on θ_e or ϕ_e , we see that

$$\psi_{n00}(r_e, \theta_e, \phi_e)|_{r_e=0} = \psi_{n00}(0, 0, 0) \quad (7.7)$$

is a constant and can be pulled out of the integral. We are left with

$$\langle \hat{\mathbf{M}}(0, \mathbf{r}_e) \rangle = \langle \boldsymbol{\mu}_S \rangle |\psi_{n00}(0)|^2 = -g_s \mu_B \langle \mathbf{S} \rangle |\psi_{n00}(0)|^2. \quad (7.8)$$

Generalizing this process slightly to allow for the dot product of the two spin operators, the expectation value of the interaction energy between the nuclear dipole and the electron spin can now be written as

$$E_{\text{hf}} = - \langle \boldsymbol{\mu}_I \cdot \mathbf{B}_S(R=0) \rangle \quad (7.9)$$

$$= - \frac{2\mu_0}{3} \langle \boldsymbol{\mu}_I \cdot \hat{\mathbf{M}}(0, \mathbf{r}_e) \rangle \quad (7.10)$$

$$= - \frac{2\mu_0}{3} \langle \boldsymbol{\mu}_I \cdot \boldsymbol{\mu}_S \delta(\mathbf{r}_e) \rangle \quad (7.11)$$

$$= - \frac{2\mu_0}{3} (-g_s \mu_B) (g_I \mu_N) \langle \mathbf{I} \cdot \mathbf{S} \rangle |\psi_{n00}(0)|^2 \quad (7.12)$$

For hydrogenic electrons in s states, we have

$$|\psi_{n00}(0)|^2 = \frac{Z^3}{n^3 \pi a_0^3} \quad (7.13)$$

and

$$\begin{aligned} \langle \mathbf{I} \cdot \mathbf{S} \rangle &= \left\langle \frac{1}{2} (\mathbf{F}^2 - \mathbf{I}^2 - \mathbf{S}^2) \right\rangle \\ &= \frac{1}{2} (F(F+1) - I(I+1) - S(S+1)) \end{aligned} \quad (7.14)$$

where $\mathbf{F} \equiv \mathbf{S} + \mathbf{I}$ is the new, total angular momentum operator and F is the total angular momentum quantum number. The hyperfine shift of the state with total angular momentum F is traditionally written as

$$E_{\text{hf}}(F) = A_{\text{hf}} \langle \mathbf{I} \cdot \mathbf{S} \rangle \quad (7.15)$$

$$= \frac{A_{\text{hf}}}{2} (F(F+1) - I(I+1) - S(S+1)) \quad (7.16)$$

and we find that

$$A_{\text{hf}}(n) = g_s g_I \mu_B \mu_N \mu_0 \frac{2Z^3}{3\pi a_0^3 n^3}. \quad (7.17)$$

Using

$$R_\infty = \frac{e^2}{8\pi\epsilon_0 a_0 \hbar c}, \quad (7.18)$$

and

$$\frac{1}{a_0} = \frac{mc\alpha}{\hbar} \quad (7.19)$$

we can write this as

$$A_{\text{hf}}(n) = g_s g_I \frac{4}{3} \frac{m_e}{m_p} \alpha^2 \hbar c R_\infty \left(\frac{Z}{n} \right)^3 \quad (7.20)$$

21 centimeter line in hydrogen

The hyperfine transition in the ground state of hydrogen occurs between the $F = 1$ (upper) and $F = 0$ (lower) manifolds and famously has a wavelength of $\lambda \approx 21$ cm. More precisely, it is known from experiments³ that the frequency of this transition is [4, 2]

$$f_{21\text{ cm}} = 1.420\,405\,751\,766\,7(10)\text{ GHz}. \quad (7.21)$$

From Eq. (7.20), with the correction for finite nuclear mass, we estimate

$$f = \frac{A_{\text{hf}}(1)}{h} = 1.422\,0\text{ GHz} \quad (7.22)$$

which gets it to better than three digits.

³This number is from the 1970s, so I wonder if anyone has done better since then. Maybe not, since this is sub-milihertz and it would be painful.

7.2.2 The nucleus creates a magnetic field at the electron

The magnetic field at position $\mathbf{r}_e$ from a nuclear magnetic dipole ($M1$) moment $\boldsymbol{\mu}_I$ at the origin can be written as

$$\mathbf{B}_I(\mathbf{r}) = -\frac{\mu_0}{4\pi} \left[\frac{\boldsymbol{\mu}_I - 3\hat{\mathbf{e}}_{r_e}(\hat{\mathbf{e}}_{r_e} \cdot \boldsymbol{\mu}_I)}{r_e^3} - \frac{8\pi}{3} \boldsymbol{\mu}_I \delta(\mathbf{r}_e) \right]. \quad (7.23)$$

Since we are going to be taking an expectation value of the magnetic field over the spatial wavefunction of the electron, we will remind ourselves that this is an operator by writing $\hat{\mathbf{r}}_e$ instead of $\mathbf{r}_e$ just to keep it straight – keep in mind that the unit vector is denoted differently ($\hat{\mathbf{e}}_{r_e}$).

The interaction energy between this field and the electron spin magnetic moment $\boldsymbol{\mu}_S$ will be described by the expectation value of the Hamiltonian $H = -\boldsymbol{\mu}_S \cdot \mathbf{B}_I(\hat{\mathbf{r}}_e)$:

$$\begin{aligned} E_{\text{hf}} &= \frac{\mu_0}{4\pi} \left\langle \frac{\boldsymbol{\mu}_S \cdot \boldsymbol{\mu}_I - 3(\hat{\mathbf{e}}_{r_e} \cdot \boldsymbol{\mu}_S)(\hat{\mathbf{e}}_{r_e} \cdot \boldsymbol{\mu}_I)}{r_e^3} - \frac{8\pi}{3} (\boldsymbol{\mu}_S \cdot \boldsymbol{\mu}_I) \delta(\hat{\mathbf{r}}_e) \right\rangle \\ &= -g_s \mu_B g_I \mu_N \frac{\mu_0}{4\pi} \left[\left\langle \frac{\mathbf{S} \cdot \mathbf{I}}{r_e^3} \right\rangle - 3 \left\langle \frac{(\hat{\mathbf{e}}_{r_e} \cdot \mathbf{S})(\hat{\mathbf{e}}_{r_e} \cdot \mathbf{I})}{r_e^3} \right\rangle - \frac{8\pi}{3} \langle (\mathbf{S} \cdot \mathbf{I}) \delta(\hat{\mathbf{r}}_e) \rangle \right] \end{aligned}$$

Since $\mathbf{S}$ and $\mathbf{I}$ are not functions of position $\mathbf{r}_e$, we can separate the calculation of expectation values of functions of only $\mathbf{S}$ and $\mathbf{I}$ (which take place in the non-spatial subspace spanned by $|SI, F M_F\rangle$ state vectors) from the terms that depend on $\mathbf{r}_e$. In the general case, this separation may not be helpful since the states of interest may no longer have individual components like M_L and M_S as good quantum numbers, which would mean we'd end up with a sum over all possibilities for those. However, since we are concentrating on the $L = 0$ case, we happen to know that $M_L = 0$ for every term, and this allows us to factor the state vector into a spatial part ($|n, L M_L\rangle = |R_{nL}\rangle |L M_L\rangle$, where the spatial wavefunction is $\psi_{n,L,M_L}(\mathbf{r}_e) = \langle \mathbf{r}_e | n, L M_L \rangle$) and a spin(s) part ($|SI, F M_F\rangle$, which has no spatial representation)⁴:

$$|n, LSI, F M_F\rangle = |n, L = 0, M_L = 0\rangle |SI, F M_F\rangle.$$

For example, the expectation value of a product of a function of spin operators with a function of the electron position operator would take the

⁴The careful reader may see here that we are defining $\mathbf{F} = \mathbf{S} + \mathbf{I}$ (and avoiding introducing $\mathbf{J}$ altogether) since $L = 0$.

form

$$\begin{aligned}
& \langle g(\mathbf{S}, \mathbf{I}) \times f(\hat{\mathbf{r}}_e) \rangle \\
&= \langle SI, F M_F | g(\mathbf{S}, \mathbf{I}) | SI, F M_F \rangle \times \langle n, 0 0 | f(\hat{\mathbf{r}}_e) | n, 0 0 \rangle \\
&= \langle SI, F M_F | g(\mathbf{S}, \mathbf{I}) | SI, F M_F \rangle \\
&\quad \times \int_0^{2\pi} d\phi_e \int_0^\pi d\theta_e \sin(\theta_e) \int_0^\infty dr_e r_e^2 \psi_{n,0,0}^*(\mathbf{r}_e) f(\mathbf{r}_e) \psi_{n,0,0}(\mathbf{r}_e).
\end{aligned}$$

In this way, the $\mathbf{r}_e$ dependence of the third term picks out the probability density at the nucleus,

$$\langle \delta(\hat{\mathbf{r}}_e) \rangle = |\psi_{n00}(0)|^2.$$

We are left with

$$\begin{aligned}
E_{\text{hf}} = & -g_s \mu_B g_I \mu_N \frac{\mu_0}{4\pi} \left[\langle \mathbf{S} \cdot \mathbf{I} \rangle \left(\left\langle \frac{1}{r_e^3} \right\rangle - \frac{8\pi}{3} |\psi_{n00}(0)|^2 \right) \right. \\
& \left. - 3 \left\langle \frac{(\hat{\mathbf{e}}_{r_e} \cdot \mathbf{S})(\hat{\mathbf{e}}_{r_e} \cdot \mathbf{I})}{r_e^3} \right\rangle \right].
\end{aligned}$$

There is more than one approach to evaluating the numerator of the last term above (*viz.* $(\hat{\mathbf{e}}_{r_e} \cdot \mathbf{S})(\hat{\mathbf{e}}_{r_e} \cdot \mathbf{I})$), but the following procedure makes it fairly clear what is happening. We will proceed by breaking this up into vector components,

$$\begin{aligned}
(\hat{\mathbf{e}}_{r_e} \cdot \mathbf{S})(\hat{\mathbf{e}}_{r_e} \cdot \mathbf{I}) &= (\hat{\mathbf{e}}_{r_e} \cdot \hat{\mathbf{e}}_x)^2 S_x I_x + (\hat{\mathbf{e}}_{r_e} \cdot \hat{\mathbf{e}}_y)^2 S_y I_y + (\hat{\mathbf{e}}_{r_e} \cdot \hat{\mathbf{e}}_z)^2 S_z I_z \\
&\quad + (\hat{\mathbf{e}}_{r_e} \cdot \hat{\mathbf{e}}_x)(\hat{\mathbf{e}}_{r_e} \cdot \hat{\mathbf{e}}_y) (S_x I_y + S_y I_x) \\
&\quad + (\hat{\mathbf{e}}_{r_e} \cdot \hat{\mathbf{e}}_x)(\hat{\mathbf{e}}_{r_e} \cdot \hat{\mathbf{e}}_z) (S_x I_z + S_z I_x) \\
&\quad + (\hat{\mathbf{e}}_{r_e} \cdot \hat{\mathbf{e}}_y)(\hat{\mathbf{e}}_{r_e} \cdot \hat{\mathbf{e}}_z) (S_y I_z + S_z I_y) \\
&= \frac{x_e^2}{r_e^2} S_x I_x + \frac{y_e^2}{r_e^2} S_y I_y + \frac{z_e^2}{r_e^2} S_z I_z + \frac{x_e y_e}{r_e^2} (S_x I_y + S_y I_x) \\
&\quad + \frac{x_e z_e}{r_e^2} (S_x I_z + S_z I_x) + \frac{y_e z_e}{r_e^2} (S_y I_z + S_z I_y)
\end{aligned}$$

Our factorization of the state vector allows us to split up the expectation values of each of these terms into a part that cares about the electronic- and nuclear-spin components (such as S_y and I_x , which is contained in $|SI, F M_F\rangle$), a part that cares about the orbital wavefunction of the electron (such as $\frac{x_e^2}{r_e^2} = \sin^2(\theta_e) \cos^2(\phi_e)$, which is contained in $|L = 0, M_L = 0\rangle$),

and a part that cares about the radial wavefunction (r_e^{-3} , which is contained in $|R_{n,L}\rangle$). We can now calculate these matrix elements directly.

For example, for the x^2 term,

$$\begin{aligned}
\left\langle \frac{x_e^2}{r_e^3} S_x I_x \right\rangle &= \langle 00 | \sin^2(\theta_e) \cos^2(\phi_e) | 00 \rangle \langle SI, F M_F | S_x I_x | SI, F M_F \rangle \\
&\quad \times \langle R_{n,0} | r_e^{-3} | R_{n,0} \rangle \\
&= \frac{1}{4\pi} \int_0^{2\pi} d\phi_e \int_0^\pi d\theta_e \sin^3(\theta_e) \cos^2(\phi_e) \langle S_x I_x \rangle \left\langle \frac{1}{r_e^3} \right\rangle \\
&= \frac{1}{3} \langle S_x I_x \rangle \left\langle \frac{1}{r_e^3} \right\rangle.
\end{aligned}$$

The orbital expectation values of the y_e^2 and z_e^2 terms likewise evaluate to $\frac{1}{3}$.

The orbital expectation value of the remaining terms evaluates to zero. One way to see this is to note that cross-term products of $\frac{x}{r}$, $\frac{y}{r}$, and $\frac{z}{r}$ (such as $\frac{xy}{r^2}$) are proportional to $Z_{L,M}^{c/s}(\hat{\mathbf{r}})$ with $L > 0$. Since these can be written as linear combinations of $Y_{LM}(\boldsymbol{\omega})$ with $L > 0$, they are orthogonal to $Y_{0,0}(\boldsymbol{\omega})$ since $\int d^2\boldsymbol{\omega} Y_{L'M'}^*(\boldsymbol{\omega}) Y_{LM}(\boldsymbol{\omega}) = \delta_{L'L} \delta_{M'M}$. (Given that $|Y_{00}(\boldsymbol{\omega})|^2 \propto Y_{00}(\boldsymbol{\omega})$, the expectation value $\langle 00 | Y_{LM} | 00 \rangle$ is proportional to the overlap integral with $Y_{00}(\boldsymbol{\omega})$, which vanishes.)

All of this allows us to write the expectation value of the hyperfine energy in the form

$$\begin{aligned}
E_{\text{hf}} &= -g_s \mu_B g_I \mu_N \frac{\mu_0}{4\pi} \left[\langle \mathbf{S} \cdot \mathbf{I} \rangle \left(-\frac{8\pi}{3} |\psi_{n00}(0)|^2 \right) \right] \\
&= g_s g_I \mu_B \mu_N \mu_0 \frac{2Z^3}{3\pi a_0^3 n^3} \langle \mathbf{S} \cdot \mathbf{I} \rangle.
\end{aligned} \tag{7.24}$$

We therefore find the same expression we got by the other approach, which had to be the case if the two were equivalent. In both cases, we see that this effect, which is known as the “Fermi contact interaction,” comes entirely from the overlap between the electron wavefunction and the nucleus. Later, we will discuss the other hyperfine terms that can appear for states with $L \neq 0$, but the basic idea of splittings due to the interaction between the nucleus and the electron (beyond the monopole interaction) will be generalizations of the ideas we investigated here.

7.2.3 The hyperfine interaction is typically weaker the spin-orbit interaction

Armed with an expression for the magnitudes of the spin-orbit and hyperfine interactions, we can compare them and develop a strategy for applying them in atoms where both are nonzero. For both to be nonzero, of course, we require $L \neq 0$, and we will assume without proving it yet that our expression for s states sets a limit for the hyperfine interaction strength even for states with $L \neq 0$.

The ratio of the fine-structure splitting coefficient to the hyperfine splitting coefficient⁵ in hydrogen scales like

$$\frac{\Delta E_{\text{SO}}}{\Delta E_{\text{hf}}} \approx \frac{Z^4 \frac{1}{2} (g_s - 1) \alpha^2 \hbar c R_\infty}{Z^3 g_s \frac{4}{3} \frac{m_e}{m_p} \alpha^2 \hbar c R_\infty} \approx Z \frac{3}{16} \frac{m_p}{m_e} \approx 350. \quad (7.25)$$

So we expect the fine structure splitting to be 2-4 orders of magnitude larger than the hyperfine splitting.

7.3 Interval Rules and State multiplicities

In our investigations of both the spin-orbit and hyperfine effects, we found that the addition of two angular momenta ($\mathbf{S} + \mathbf{L} = \mathbf{J}$ for spin-orbit coupling, $\mathbf{S} + \mathbf{I} = \mathbf{F}$ for the hyperfine interaction in s -state atoms) to make a third led to a collection of levels that were split by some energy that was considered small compared to the other energy scales in the unperturbed Hamiltonian. In particular, we treated these interactions as perturbative and argued that it would be helpful to retain the angular momentum quantum numbers of the constituent momenta (S and L or S and I) as labels.

To generalize this slightly, if there is a perturbation that is proportional to $\mathbf{J}_1 \cdot \mathbf{J}_2$, it is likely that the two momenta being added are $\mathbf{J}_1$ and $\mathbf{J}_2$ to make $\mathbf{J}$, and we argued that the eigenstates of $\mathbf{J}^2$ and J_z could also carry those other labels just to help us remember the basic content of the state,

$$|J_1 J_2, J M_J\rangle. \quad (7.26)$$

More than being just a convenience, this can become especially important in cases where there are multiple states with the same eigenvalue of $\mathbf{J}^2$ but some

⁵I am saying “coefficient” here to mean that I am removing all dependence on quantum numbers and nuclear g factors.

other quantum numbers that are different (and lead to different energies), such as $^2S_{1/2}$ and $^2P_{1/2}$.

We find that the $\mathbf{J}_1 \cdot \mathbf{J}_2$ interaction leads to energy shifts that are given by some constant times $\langle \mathbf{J}_1 \cdot \mathbf{J}_2 \rangle$. We can write this as

$$\begin{aligned} E_{\mathbf{J}_1 \cdot \mathbf{J}_2} &= C \langle \mathbf{J}_1 \cdot \mathbf{J}_2 \rangle \\ &= \frac{C}{2} (J(J+1) - J_1(J_1+1) - J_2(J_2+1)). \end{aligned} \quad (7.27)$$

We can calculate the energy difference between a state with quantum number J and a state with quantum number $J' = J - 1$,

$$\begin{aligned} \Delta E_{\mathbf{J}_1 \cdot \mathbf{J}_2} &= \frac{C}{2} (J(J+1) - (J-1)J) \\ &= C J. \end{aligned} \quad (7.28)$$

This result, Eq. (7.28), is known as an *interval rule*, and it's a nice simplification compared to trying to remember Eq. (7.27) as long as you're only interested in the splitting between levels that are the same but for the fact that their total angular momentum quantum numbers differ by precisely 1.

Further, just as the case was with the hyperfine interaction, we can repeat the same procedure for determining which eigenvalues J will be allowed. To put this explicitly, we can follow a three-step procedure:

1. Start by calculating the maximum allowed value of J ,

$$J_{\max} \equiv J_1 + J_2 \quad (7.29)$$

and add this to your list of allowed values of J .

2. Subtract 1 from the lowest value of J that you have determined is allowed. Compare this to the minimum allowed value of J ,

$$J_{\min} \equiv |J_1 - J_2|, \quad (7.30)$$

and add it to the list of allowed values of J as long as it is greater than or equal to $J_{\min}$.

3. Repeat Step 2 until you can no longer subtract 1 while keeping the result greater than or equal to $J_{\min}$.

All of this, of course, is equivalent to making a list of every J from $J_{\max}$ to $J_{\min}$ in steps of $\Delta J = 1$.

Given that the hyperfine interaction was estimated to be a few orders of magnitude weaker than the spin-orbit interaction, we can treat the spin-orbit interaction as a perturbation on the Coulomb interaction (between the nucleus and the electron) and then treat the hyperfine interaction as a perturbation on the spin-orbit interaction. The states will be built as follows:

1. Solve the Coulomb Hamiltonian to get states $|n, L M_L\rangle$. Good quantum numbers at this level are $\{n, L, M_L, e/o\}$
2. Add electron spin: Treat the spin-orbit interaction as a perturbation that yields eigenstates $|n, LS, J M_J\rangle$. Good quantum numbers at this level are $\{J, M_J, e/o\}$, with approximately-good quantum numbers $\{n, S, L\}$.
3. Add nuclear spin: Treat the hyperfine interaction as a perturbation that yields eigenstates $|n, LSJI, F M_F\rangle$. Good quantum numbers at this level are $\{F, M_F, e/o\}$, with approximately-good quantum numbers $\{n, S, L, J, I\}$ ⁶.

This prescription is not the only possibility. If the orbital hyperfine interaction is stronger than the spin-orbit interaction (which is rare, but possible), the nuclear spin could couple with the electron orbital angular momentum to make some combined angular momentum $\mathbf{G} \equiv \mathbf{I} + \mathbf{L}$ with states $|n, LI, G M_G\rangle$ (where $\{n, L, I\}$ would now be approximately good quantum numbers). The spin could then have a weaker interaction with $\mathbf{G}$ to make $\mathbf{F} = \mathbf{G} + \mathbf{S}$, for which the states would be $|n, LIGS, F M_F\rangle$.

In fact, any basis could be used for calculations and you'd always get the correct answer as long as you remember that the only strictly-good quantum numbers are $\{F, M_F, e/o\}$. The point of writing the couplings by starting

⁶You may be wondering why I am calling I an approximate, as opposed to exact, quantum number for an atom. The reason is that, strictly speaking, the hyperfine interaction can mix in excited nuclear states for a composite nucleus and a multi-electron atom. Of course, those excited nuclear states are *extremely* high in energy (except for ²²⁹Th) and we will generally assume that I is a completely good quantum number in practice. Mathematically, however, we treat it the same as the rest, and there's no reason to mess up the symmetry of our description by promoting that particular angular momentum to exact status.

with the strongest and working you way down is simply to define the approximate quantum numbers in a way that makes them as good as possible.

In this course, we are typically going to ignore these other choices and we will assume that our good eigenstates are characterized by $\{n, L, S, J, I, F, M_F, e/o\}$.